COMPSCI 713: AI Fundamentals

Lecture 12: Embodied AI and AI Teams

Polly, BigDog, and robot soccer

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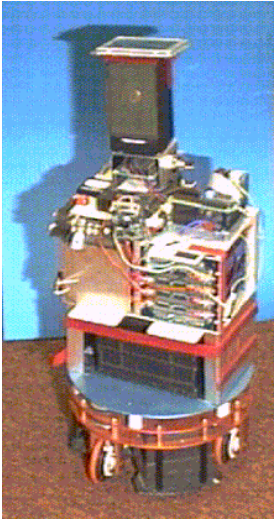
Overview

- We begin with **embodied AI**: systems that control a physical body and must react to sensors in real time.
- This naturally leads to **situated reasoning**, layered control, and environment-specific shortcuts.
- Core examples: **Polly**, **BigDog**, and autonomous robots for **Mars exploration**.
- Then we move from single robots to **teams and swarms**: joint goals, flocking, and robot soccer.
- Main thread: intelligence is often about *acting robustly in the world*, not only about abstract reasoning.

This version keeps the original content, but organizes it as a smoother two-part story: embodied autonomy first, coordinated multi-agent behavior second.

Polly

- First visually navigated system to move at roughly **animal speed** (about 1 m/s) back in 1993.
- Demonstrated Rodney Brooks' idea of using **simplifying assumptions** to build a robot that works well in a specific context.
  - Much like a simple animal, Polly did not need a high-level understanding of the world to do its job.
- Polly gave tours of the MIT AI Lab's 7th floor for several years.
  - Played canned speech at designated positions.



Polly's assumptions

- **Environment:**
  - uncluttered office,
  - corridors,
  - uniform un-patterned carpet.
- This setting enabled remarkable computational shortcuts:
  - **Carpet / object detection:** if it is not carpet, it is probably an obstacle; stuff with patterns are obstacles
  - **Ground-plane constraint:** Objects rest on the flat floor, so objects that appear higher in the image are further away.
  - **Corridor geometry:** And narrow corridors constrain where in the visual field more distant landmarks can appear, reducing the search space

Design principle

Do not solve the hardest possible vision problem if the environment lets you solve an easier one.

Polly's visual system

- Took a **64** $\times$ **48** image every **66 ms**.
  - Extremely low resolution – but therefore cheap to compute on with low power.
- Each frame produced a set of visual **precepts** (mostly Booleans), including:

`open-left?`	`open-region?`	`person-direction`
`open-right?`	`blind?`	`wall-ahead?`
`blocked?`	`light-floor?`	`wall-far-ahead?`
`left-turn?`	`dark-floor?`	`vanishing-point`
`right-turn?`	`person-ahead?`	`farthest-direction`

- Example operations:
  - Radial depth map (like you might get from sonar) ground plane constraint lets us use height as depth
  - Edge detection assumes strong straight corridor edges and prioritizes finding them

Polly navigation and control

- Stored a library of frames (records) describing how landmarks looked from different directions and distances.
  - Low resolution made those visual searches surprisingly efficient.
- Used rough **rotational odometry**: it tracked heading from how much it had turned left and right.
- Also recognized **districts** to recover localization.
  - For example, if its going east and sees a left turn ahead it must be in the southern corridor and can set its y coordinate to 10 (helps with recovering from getting lost)
- Visitors requested a tour by waving a foot.
  - Polly system had no speech recognition and the camera only looked at the level of peoples feet/legs

Navigation here is appearance-based and pragmatic, not a full symbolic world model.

Predecessor to Polly: Allen

- Allen (1986) demonstrated Brooks' idea of a **layered control system**.
- Used sonar sensors to get ranges to obstacles

- **Level 0 – Avoid:** veer away from nearby obstacles; stopping force scaled roughly with inverse-square distance.
- **Level 1 – Wander:** choose a random direction and follow it for about 10s.
- **Level 2 – Explore:** head in a direction of wide-open space.

- The robot combined forces from all three levels to choose a direction.
- Result: robust wandering with no master plan, no rich knowledge, and very simple programming for more "intelligent" robots.

Predecessors to Polly: Squirt, Genghis, Herbert

- **Squirt:** a tiny robot (about 1 cubic inch) that sought dark places and sat still in corners.
  - Layers a bit like Allen, but with difference sensors
  - Fundamental behaviour was to seek dark places and then sit still in a corner
  - Upper behaviour was to stay still for several seconds after a loud noise, then move quickly directly toward it
  - Then the first behaviour would kick in and it would hide, but now closer to "where the action is"
- **Herbert:** equipped with an arm and programmed to steal empty soda cans.
- **Genghis:** a six-legged robot showing **loose control**: legs were not explicitly coordinated, yet it still scrambled around and followed infrared sources.

BigDog

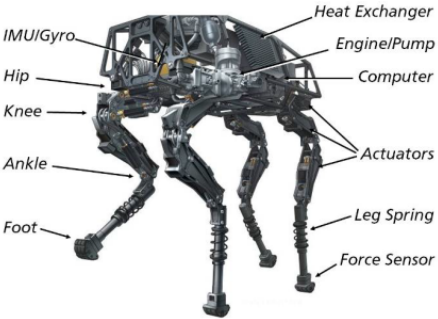
- What if a robot could move over **rough terrain**?
- What if it used **legs**, like many animals, rather than wheels or tracks?
- The U.S. military, via DARPA funding, considered this a serious design direction.
- BigDog became a famous rough-terrain quadruped prototype.

Demo video:
`https://www.youtube.com/watch?v=cNZPRsrwumQ`

Physical architecture

- Two-stroke water-cooled internal combustion engine (like a cheap old motorcycle!) powered a hydraulic pump.
- High-pressure oil drove servo valves and hydraulic cylinders for the leg actuators.
- Sensors measured joint position and force in the legs.
- Inertial sensors measured body angle and acceleration.
- An onboard computer ran a **two-level control system**.

Mass: about **109 kg**.



Walking on legs

- Low-level control manages the position and forces of individual joints.
- High-level control coordinates legs to regulate body speed, attitude (pitch, yaw, roll), and altitude (e.g. crouched or standing tall).
- Individual legs generate ground reactions directed toward the hip.
- BigDog can switch among **crawl**, **walk**, and **trot** gaits.
- Trot is used for speed; crawl helps on steep or slippery ground.
- The controller also adjusts torso pitch for climbing and descending.

Dynamic balancing

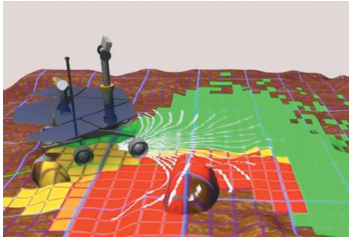
- It is not surprising that learning-based algorithms can be trained to control legs.
  - This belongs to the broader class of balance problems: earlier in the course we saw NEAT solve a two-pole balancing task in 2D.
- One-legged hopping robots make this class of control problem especially easy to visualize. BigDog lab predecessors: `https://www.youtube.com/watch?v=XFXj81mvInc`
- In many lab videos, a **tether** is visible because power supply and/or computation remained offboard to keep the prototype light and cheap.

BigDog: high-level behaviours and fate

- Stereo vision and LIDAR let the robot estimate terrain shape ahead and follow a leader.
- Outside follow mode, it often seemed to be "driven" a bit like a drone.
- The prototype was **very loud**.
  - The military eventually lost interest, but that does not mean the idea is a dead end for all applications.
- That doesnt mean its a dead-end for every application, or forever
  - some researchers suggested a four-stroke engine, muffler or hybrid operation would be options, but all that would make it heavier; eventually youd have something more like a car!
  - And of course batteries have gotten better since then - but plugging in to recharge on a battlefield is a huge limitation.
  - Still a promising direction for **search-and-rescue**, **hazardous-site inspection**, and **mine clearing**.

Robots on Mars

- Some degree of autonomous control is essential for Mars rovers.
  Why?
- A lot of elements similar to the prototypes weve seen
- Stereo cameras can build a terrain map.
- Wheel odometers estimate motion – but are unreliable when slipping on sand.
- Inertial sensors estimate pitch and roll.
- Heading can be estimated from sun position and time.
- The rover can plan a path, adapt it dynamically, and respect human-labeled no-go areas.



Mars rovers repeat the same theme: embodiment plus clever approximations under harsh constraints.

Space travel is a harsh deal

- Rover engineering is often maddeningly conservative.
  - Hardware must survive radiation and extreme temperatures.
  - Power is precious: for solar missions, dust on panels can end the mission.
  - Rovers move **very slowly**; tipping over would be disastrous.
- Some researchers suggest releasing swarms of Genghis-like bug robots.
- The **Ingenuity** helicopter was a calculated risk that paid off: planned for 5 flights, it completed 72.

Conclusion to part 1

- Embodied AI platforms are becoming increasingly competent at moving through the world.
  - Legged systems may eventually reach places wheels cannot – perhaps cluttered homes, not just neat hospital corridors.
  - Soft computing, machine learning, and task-specific shortcuts help with low-level motion and navigation.
- These platforms can support higher-level applications such as classical planning, CNN vision, speech recognition, and LLM-based query-response.
- Then again, be sceptical of the hype!
  - So many videos show amazing robot behaviour that doesnt turn into viable products but maybe soon!

Changi Airport, Singapore, 2023

Still a ways to go before robots match what animals can do

- Animals still dominate on agility, robustness, and energy efficiency.
- Biological systems remain a benchmark as well as an inspiration.
- The gap between competent demos and all-purpose competence is still large.

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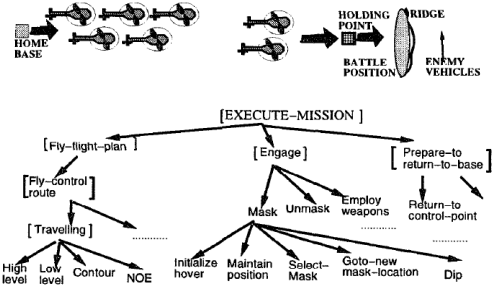
Social animals

- Many intelligent animals – especially our close relatives – are highly social.
- Some collective systems, such as bees, show rich organization even if individuals are not especially smart in isolation.
- Strong AI should be able to produce something analogous.

Joint reasoning

- Tambe (1997) studied how to do "classic" AI planning with a physically distributed team of agents.
- One motivating domain was a squadron of **attack helicopters**.
- A brittle team plan can fail badly; for example, if a scout is destroyed and everyone else waits forever for the all-clear.
- Adding one extra rule may patch one failure, but ad hoc patches proliferate and other failures remain hidden.



STEAM (a Shell for TEAMwork)

- Core idea: commitment to a **joint persistent goal (JPG)**.
  - Team members keep pursuing the goal unless they learn it is **Achieved**, **Unachievable**, or **Irrelevant**. (A v U v I)
- If one agent privately concludes (A v U v I), it exits the JPG.
  - But the JPG is replaced by a commitment to have new **mutual belief** for the team (either A, or U, or I)
- So STEAM is fundamentally a commitment to **communicate**, not just to act.
- Tambe also compared communication strategies, since naive communication can generate too many messages.

Example

If a team member detects a surface-to-air missile battery and the mission becomes **unachievable**, it should not simply fly home without explanation.

“Elephants don’t play chess”

- Brooks’ famous paper (1990) a famous paper "Elephants dont play chess"
  - He actually doesnt say much materially about elephants
  - But the point is that intelligence doesnt start with chess
  - As such, maybe tools such theorem provers arent the right basis for all of general AI
- Brookes notes that evolution through most of Earth history was concerned with simple life forms (a billion years plus)
  - And then from the Cambrian Explosion we see larger animals for a few hundred million years
  - Its only been a brief moment relatively that human-level intelligence has been at play (around a million years)
- So general AI should also study **simple intelligent behavior**: cockroaches, rats, locomotion, and social coordination.
- The broader message is methodological: **build upward from situated competence**, not downward from elite symbolic performance.

This provides the conceptual bridge from embodied robots to flocking and teamwork.

Murmurations of starlings

- Mesmerizing patterns can emerge from many thousands of birds moving together.
- Which bird decides where to go?
- How is the "plan" communicated through the flock?
- These questions motivate distributed coordination without central control.

Video: `https://www.youtube.com/watch?v=V4f_1_r80RY`

Three simple rules (Reynolds, 1987)

R1: Collision avoidance

Do not come within a minimum distance of nearby flockmates.

R2: Flock centering

Stay close to flockmates rather than drifting away from the group.

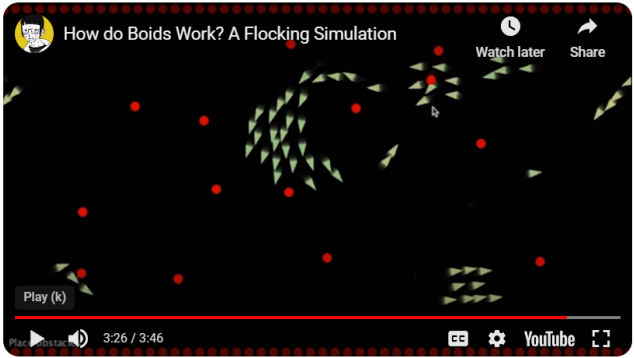
R3: Velocity matching

Align your motion with nearby agents. Velocity is a vector: it includes both speed and direction.

- This is a form of **agent-based modelling**: global behavior emerges from local interactions.
- Similar modeling ideas appear in epidemiology and social-network dynamics.

“Boids” (bird-droids) simulator

- Reynolds’ rules can already generate striking flock-like behavior.
- The simulator is a clean demonstration of **emergence**: realistic patterns from simple local rules.
- It also helps us attribute different visible behaviors to different rules.



Demo: `https://www.youtube.com/watch?v=QbUPfMXXQIY`

The model in more detail

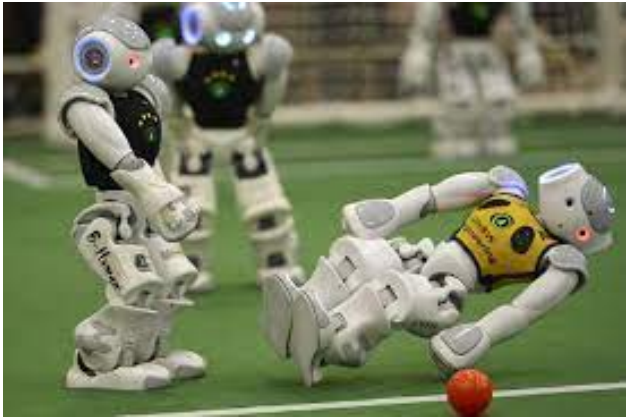
- Hermellin and Michel's (2017) implementation uses **5 parameters** and **3 agent attributes**.
  - **Parameters:** field of view, minimum separation distance, cohesion threshold, maximum speed, maximum rotation.
  - **Agent attributes:** heading, speed, and nearest-neighbor list.
- Rules are implemented in terms of these parameters
  - Example of **R2**: when several agents are close – but not too close – an agent adjusts its heading toward the mean heading of visible neighbors.
  - Example of **R1**: if an agent is too close to another, it looks to see if its heading and its neighbours heading leads to a collision; if so, rotates to avoid
- Small implementation choices can change the emergent flocking behavior a lot.

Robot soccer

- Robot soccer is a natural testbed for embodied multi-agent AI.
- It combines perception, locomotion, communication, coordination, and strategy.
- Competition footage makes the coordination challenge very concrete.

World Cup 2023: `https://www.youtube.com/watch?v=3kdWkXZYEV8`

Photo credit in original slides: Al Jazeera



How can the players coordinate?

- The answer depends strongly on the rules of the league.
  - Can players access a **shared data model** over WiFi?
  - Is there an **overhead camera view**?
  - Can robots signal each other directly, or talk to a control computer, or only use body signals?
- Perception from each robot can be very limited: walking platforms are unstable and their field of view is narrow.
  - In that case, each robot may only maintain a rough **probabilistic model** of the game state.

Aspects of robot soccer player coordination

- **Collective behaviors:**
  - e.g., passing strategies,
  - candidate passing points derived from interception prediction,
  - scoring those passing points by position and tactical value.
- **Positioning strategies**
  - choose formations that create both attacking and defensive opportunities.
- **Role-based strategies**
  - assign roles dynamically based on the game situation.
- Position- and role-based methods work best when teammates perceive the situation in a sufficiently similar way.

From this module

- We have now seen: rule-based expert systems, induced decision trees and forests, fuzzy logic, Naïve Bayes classifiers, genetic algorithms, walking robots, and team behavior.
- Together these methods let us:
  - represent and operationalize specialized expert knowledge,
  - create controllers using heuristics for specific-purpose devices,
  - build classifiers from labeled examples (for example, likely spam detection),
  - refine controllers from feedback, often with many simulations,
  - coordinate teams through joint goals, flocking rules, and role selection.

Conclusion

- Many important AI successes before LLMs were easy to overlook because they lived inside embedded controllers.
  - E.g. smart controllers embedded in so many systems: washing machines, vacuum cleaners, cars and planes (e.g. anti-lock braking systems, ABS; autopilots) even if a human still drives at a high level
- Many AI approaches borrow ideas from nature: Genetic algorithms, and neural networks.
- Embodied robots have not yet reached science-fiction dreams, but the core building blocks already exist for wider use in specific applications.

References (part 2)

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Thank you!

Welcome Yueying for next part!